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Agentic Verification of Municipal Incident Reports through Grounded Dialogue and Multimodal Evidence

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Abstract

Municipal call centers receive incident reports whose reliability ranges from urgent and accurate to vague, duplicated, mistaken, or deliberately false. We describe an agentic AI system for triaging non-emergency reports by combining adaptive questioning, consent-based location evidence, street-level imagery, public service-request history, and external news or official feeds. The system is framed around a New York City 311 pothole-reporting scenario, where a citizen calls about a road hazard that may require Department of Transportation action. Rather than treating a caller as truthful or deceptive from language alone, the system builds a grounded claim graph and estimates whether the report is actionable, uncertain, or likely abusive. The design emphasizes low-friction questioning, evidence provenance, human review, and safety-first escalation when injury or imminent danger is possible. We argue that agentic verification can reduce prank-call load while preserving access for genuine reporters, provided it is deployed with privacy, bias, and accountability constraints.

1 Introduction

City service centers are increasingly asked to react to public reports in real time: potholes, graffiti, broken traffic signals, petty crime, vandalism, illegal dumping, street flooding, or noise. The operational problem is not simply classification. A call may be sincere but imprecise, duplicated but useful, maliciously false, or a serious emergency incorrectly routed to a non-emergency line. A rigid form or a single language-model judgment is therefore risky. The system must ask the right follow-up questions, use external tools, keep evidence separate from speculation, and route the case to the correct human or agency. We propose a work in progress work, CIVICVERIFY, an agentic architecture for verifying municipal incident reports. The agent does not decide whether a citizen is honest in a moral sense. Instead, it estimates whether a report is actionable under agency rules. Its core question is: given the call transcript, reported location, optional device location, public geographic context, service history, and corroborating public information, should the city dispatch, request more information, escalate to a human operator, merge with an existing ticket, or mark the call as likely abusive? Our design (as illustrated in figure 1) draws on tool-using and reasoning-acting language-model paradigms (Yao et al., 2023; Schick et al., 2023), retrieval-augmented generation for grounding decisions in external evidence (Lewis et al., 2020), and fact-verification work that distinguishes supported, refuted, and insufficiently supported claims (Thorne et al., 2018). We use natural language inference and consistency checks (Williams et al., 2018) not to infer deception from accent, hesitation, or affect, but to compare concrete claims across turns. The main contribution is a practical short-paper specification of an incident-verification agent for city call centers.

2 Scenario: NYC 311 Pothole Reports

New York City provides a useful example because pothole reports are accepted through NYC311 and routed to the Department of Transportation. The public NYC311 pothole page states that DOT will fill most street potholes within 15 days after a street-pothole service request is filed (NYC311, 2026). NYC Open Data also publishes service-request records for 2020 to the present, including fields such as complaint type, responding agency, status, and geographic location, while excluding personally identifying customer information (City of New York Open Data, 2026).

Consider a call-center report: “There is a huge pothole on Northern Boulevard near 80th Street in Queens, eastbound right lane, near the bus stop. Cars are swerving.” A serious call should be cap-

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Figure 1: Proposed workflow

tured quickly because road defects can cause vehicle damage and safety risks. A troll call might instead say that a "sinkhole swallowed five cars in Times Square" while refusing to provide a cross street, claiming impossible landmarks, or repeatedly changing the location. In both cases the agent must avoid overconfidence. Google Street View can help verify road geometry and nearby landmarks, but its imagery is historical and metadata may have only month or year granularity; it cannot prove a pothole exists at call time (Google Maps Platform, 2026). Thus the system treats Street View as context, not final evidence.

3 System Overview

CIVICVERIFY has six cooperating agents. The *intake agent* transcribes the call, detects incident type, and extracts a frame: location, time, hazard, severity, direction of travel, landmarks, and whether anyone is injured. The *geospatial agent* geocodes the report, compares it with optional one-time device location, checks jurisdiction boundaries, and identifies road segments. The *visual-context agent* retrieves street-level imagery and metadata for landmarks, lane structure, bus stops, storefronts, and road geometry. The *retrieval agent* queries city service-request history, official feeds, and reputable local news. The *dialogue agent* asks short follow-up questions selected by expected information gain. The *decision agent* calibrates a final routing recommendation and attaches an evidence card.

The agents communicate through a claim graph. Nodes are claims such as *incident type=pothole*, *street=Northern Boulevard*, *direction=eastbound*, *near landmark=Q32 bus stop*, and *vehicles swerving*. Edges represent support, contradiction, duplication, or missing evidence. Tool outputs are stored with provenance, timestamp, confidence, and use restrictions. This separates the model's reasoning from the facts available to the city.

4 Grounded Dialogue Policy

The dialogue policy is designed for short municipal calls. It asks only questions that can change routing. For a pothole, useful questions include: "What is the nearest cross street?", "Which direction of traffic is affected?", "Which lane is it in?", "Is there a nearby bus stop, hydrant, school, or storefront?", and "Is anyone hurt or is the lane fully blocked?" If the caller uses a mobile device, the operator may ask for a one-time location share. Refusal is not treated as deception; it only removes one evidence source.

Question selection follows a simple utility model. Let H be the current hypothesis set over routing outcomes: actionable, duplicate, insufficient information, out of jurisdiction, emergency, or likely abusive. For each candidate question q, the dialogue agent estimates

$$U(q) = \Delta I(H; q) - \lambda C(q) - \mu R(q), \quad (1)$$

where ΔI is expected information gain, $C(q)$ is caller burden, and $R(q)$ is privacy or sensitivity risk. The agent asks the highest-utility question unless a safety trigger requires immediate transfer.

Consistency is computed over semantic claims, not speaking style. If a caller first says the pothole is "in Queens by 80th Street" and later says "near 79th Street," that may be acceptable. If the caller alternates among Queens, the Bronx, and Times Square while refusing landmarks, the report receives a contradiction flag. Natural language inference models can assist with such comparisons, but behavioral testing is necessary because NLP systems often fail outside their intended distribution (Ribeiro et al., 2020). Long transcripts are summarized into claim graphs because relevant evidence can be overlooked in long contexts (Liu et al., 2024).

Signal	Example	Interpretation
Location fit	GPS near geocoded segment	Supports report presence, if consented
Street context	Reported lane and bus stop visible	Supports plausibility, not current pothole
Service history	Similar recent 311 tickets	Supports urgency or duplicate merge
Dialogue consistency	Same cross street after re-asking	Supports content reliability
Contradiction	Landmark absent from context	Raises uncertainty; ask again
Safety trigger	Injury, crash, fire, weapon, blocked route	Escalate regardless of score

Table 1: Evidence used for actionability scoring. Signals are not used to punish callers; they determine routing and review level.

5 Decision Model

The final routing score is not a single LLM label. It is a calibrated evidence model:

$$P(y = \text{actionable} \mid C, E) = \sigma(w^\top f(C, E)), \tag{2}$$

where C is the call-derived claim graph, E is external evidence, and f contains features from Table 1. The output classes are *dispatch*, *merge duplicate*, *request more information*, *human review*, *out of jurisdiction*, *emergency transfer*, and *likely abusive*. The final label is constrained by policy rules. Any injury, crash, fire, active crime, blocked critical route, or imminent danger is routed to human emergency review even when the actionability score is low. The decision card explains the recommendation in operational terms. For the serious pothole call, it might say: reported segment geocoded to Northern Boulevard eastbound; caller shared one-time location within 45 meters; street imagery confirms the described bus stop and lane layout; two similar recent 311 requests exist; no contradiction across three follow-up answers; create DOT pothole ticket and merge possible duplicates. For the troll-like call, it might say: caller gave three mutually inconsistent locations; claimed a landmark not found at the reported intersection; refused all clarifying details; no corroborating city ticket or official alert; non-emergency and no injury; send to human review before closure as likely abusive.

6 Worked Call Flow

The following call flow illustrates how the agent distinguishes a serious report from a low-value abusive one without relying on personality judgments. In the serious case, the caller reports: "There is a deep pothole on Northern Boulevard, eastbound, just after 80th Street, by the bus stop. Cars are moving left to avoid it." The intake agent extracts a pothole frame and asks two questions: "Is anyone injured or is traffic fully blocked?" and "Can you confirm the nearest cross street or landmark?" The caller answers that no one is injured, the right lane is affected, and the bus stop sign is immediately past the intersection. The caller consents to one-time location sharing. The geospatial agent finds that the device location is close to the reported segment. The visual-context agent verifies that the road segment and landmark pattern are plausible, while noting that the imagery is not current. The retrieval agent finds recent similar service requests in the same corridor. The decision agent creates a DOT pothole ticket, attaches the supporting evidence, and marks the case as a possible duplicate if an open ticket already exists. In the low-value case, the caller says: "A giant hole swallowed cars outside Times Square; send everyone now." The dialogue agent first checks for emergency indicators and asks whether anyone is injured, which lane is blocked, and the nearest cross street. The caller alternates between Times Square, Queens, and Brooklyn, refuses all landmarks, and gives claims that cannot be mapped to a single road segment. Because no injury is confirmed and the report is internally inconsistent, the agent does not dispatch automatically. It places the case in human review with a recommendation of likely abusive or insufficient information. This preserves a path for a genuine but distressed caller while preventing the model from silently closing a potentially serious report. A key detail is that the system separates *absence of evidence* from *evidence of falsity*. Missing GPS, unavailable Street View metadata, or lack of news coverage should not automatically lower a report below the action threshold. Many real potholes, graffiti tags, or broken signs are too local to appear in news feeds. The agent therefore assigns strongest weight to stable and specific claims that an operator can act on: a mappable segment, direction of travel, lane, landmark, and safety status.

7 Failure Modes and Mitigations

Several failure modes are expected. First, street-level imagery may be old, missing, or captured from the wrong side of the road. The mitigation is to treat it as a plausibility check only. Second, service-request history can amplify existing inequities: neighborhoods with more digital access may have more prior reports. The mitigation is calibration by geography and operator audit rather than a universal threshold. Third, a malicious caller can provide plausible details copied from maps. The mitigation is not aggressive interrogation, but evidence fusion: location fit, duplicate history, consistency, and human review for borderline cases. Fourth, multilingual callers or callers under stress may give inconsistent wording while reporting a real issue. The mitigation is language support, relaxed matching for near-equivalent locations, and a rule that uncertainty leads to clarification or review rather than punishment. The system should also handle category drift. A call that begins as a pothole report may become an emergency if the caller mentions a crash or injury. A vandalism report may become a safety report if broken glass blocks a school entrance. The dialogue agent therefore re-runs safety detection after each turn. The decision agent is constrained so that emergency transfer overrides all prank-call heuristics.

8 Evaluation Plan

A realistic evaluation should combine historical data, simulation, and field review. Historical 311 data can identify complaint categories, duplicate patterns, agency outcomes, and resolution status. However, historical records alone do not reveal caller intent. We therefore propose three evaluation sets. First, an offline set links calls or text submissions to later agency outcomes, such as inspector confirmation, work-order creation, or duplicate closure. Second, a synthetic adversarial set introduces prank patterns: impossible locations, changing cross streets, exaggerated claims, copied news headlines, and harassment. Third, a human-reviewed pilot set records operator decisions, agency follow-up, and user satisfaction. Metrics should reflect public-service risk. Overall accuracy is not enough. The most important metric is false dismissal of serious reports. Other metrics include time to create a ticket, number of unnecessary follow-up questions, duplicate-merge precision, human-review rate, complaint resolution latency, and calibration error. Evaluation should be stratified by borough, language, device type, accessibility needs, and neighborhood service history, because an agent that is stricter in low-income areas or with non-native speakers would create unequal access.

9 Safety, Privacy, and Governance

The system should use data minimization. One-time location sharing should be explicit, revocable, and limited to the current report. Street View and map data should be used for geography and landmark plausibility, not surveillance. Caller history should be restricted to operational abuse prevention, not broad profiling. The model must never infer truthfulness from accent, emotion, fluency, disability, or background noise. The phrase "likely abusive" should mean that the report content lacks operational support after targeted questions, not that the person is a liar.

Human oversight is required for consequential outcomes. NIST's AI Risk Management Framework emphasizes governance, documentation, transparency, accountability, and risk management for AI systems (NIST, 2023). For municipal use, this means audit logs for tool calls, clear explanations for operators, appeal or resubmission paths for citizens, periodic bias audits, and conservative escalation rules. The agent should improve the operator's evidence-gathering workflow rather than replace city responsibility.

10 Conclusion

Agentic AI can help city call centers separate serious incident reports from vague, duplicate, or abusive calls by combining grounded dialogue with external evidence. In the NYC 311 pothole scenario, the agent verifies location plausibility, asks concise clarifying questions, consults service history, uses Street View only as historical context, and produces a routing recommendation with provenance. The main design principle is safety-first actionability: when harm is possible, escalate; when evidence is incomplete, ask or review; when contradictions accumulate in a non-emergency report, reduce operational priority without punishing the reporter. This approach offers a practical path for responsible municipal AI deployment.

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